

Systematic Literature Search: Peer-Reviewed Meteorological Journal Articles Using the Exact Phrase "Spanish Plume" (published on or before 13 August 2023)

TL;DR

- A direct, independent phrase-search across the four required databases identified **7 confirmed-eligible** peer-reviewed meteorological journal articles containing the exact phrase "Spanish plume"/"Spanish plumes" in title, abstract, or body and published on or before 13 August 2023, plus **3 additional strong-candidate papers that remain "unverified"** because paywalled full text prevented confirming the phrase appears in the body rather than only in references.
- The search's independence was **materially challenged**: the Schultz et al. (2025) review articles (the very sources excluded by instruction) dominated nearly every query's results, so the search had to actively resist harvesting their reference lists; discovery was instead driven by publisher/journal phrase queries.
- **Raw database counts could not be verified** for Google Scholar, Web of Science, or AMS Journals Online because those search interfaces require authentication or block automated queries; counts are reported as "inaccessible/unverified" rather than fabricated.

Key Findings

- The confirmed-eligible corpus is small and heavily concentrated in RMetS/Wiley journals, especially *Weather*.
- Confirmed eligible papers: Sibley (2012, *Weather*); Webb & Pike (1998, *Weather*); van Delden (1998, *Meteorology and Atmospheric Physics*); Lewis & Gray (2010, *Atmospheric Research*); Lewis & Silkstone (2017, *Weather*); Holley et al. (2014, *International Journal of Climatology*); Hayward et al. (2023, *International Journal of Climatology*).
- Unverified candidates (topically Spanish-plume, phrase not confirmable in body): de Villiers (2020, *Weather*), Young (1995, *Weather*), van Delden (2001, *Atmospheric Research*).
- Notable near-misses/borderline: Morris (1986, *Meteorological Magazine*) — the phrase is in the title but *Meteorological Magazine* is a Met Office professional/technical magazine, not a peer-reviewed journal → excluded; Gray & Marshall (1998, *Weather*) — Spanish-plume-adjacent MCS climatology, phrase not confirmed.
- Both Schultz et al. (2025) reviews appeared at or near the top of essentially every query but were excluded (published in 2025, after the 13 August 2023 cut-off, and are the designated non-independent sources).

Details

1. Per-database search log

Database	Exact query string	Date searched	Raw result count
Google Scholar	"Spanish plume"	31 Jul 2026	Unverified — results page could not be retrieved (access blocked); Scholar cannot restrict to peer-reviewed or to meteorology, and returns many non-meteorological false positives (volcanic plumes, dispersion, Spanish-language items)
Wiley Online Library / RMetS journals (<i>QJRMS</i> , <i>Weather</i> , <i>Meteorological Applications</i> , <i>Atmospheric Science Letters</i> , <i>International Journal of Climatology</i>)	"Spanish plume" and "Spanish plumes"	31 Jul 2026	Unverified numeric total ; individual article records retrieved and confirmed (see eligible table)
AMS Journals Online (<i>MWR</i> , <i>WAF</i> , <i>JAS</i> , <i>BAMS</i> , etc.)	"Spanish plume"	31 Jul 2026	Unverified — search-results total not machine-readable; individual article XML pages accessible
Web of Science Core Collection	TS=("Spanish plume")	31 Jul 2026	Unverified — requires institutional authentication; not accessible

Because none of the four native database result counts could be verified, discovery relied on federated web/publisher phrase queries and direct confirmation on each publisher page. This is a significant methodological limitation (see Caveats).

2. Eligible papers (exact phrase confirmed in title/abstract/body; peer-reviewed; meteorological;

Title	Authors	Year	Journal	DOI	Found via	Phrase location
Thunderstorms from a Spanish Plume event on 28 June 2011	Sibley, A.	2012	<i>Weather</i> 67(6):143–146	10.1002/wea.1928	Wiley/RMetS	Title + body
Thunderstorms and hail on 7 June 1996: An early season 'Spanish plume' event	Webb, J.D.C.; Pike, W.S.	1998	<i>Weather</i> 53(8):234–241	10.1002/j.1477-8696.1998.tb06391.x	Wiley/RMetS	Title
The synoptic setting of a thundery low and associated prefrontal squall line in western Europe	van Delden, A.	1998	<i>Meteorology and Atmospheric Physics</i> 65(1-2):113–131	10.1007/BF01030272	Springer	Abstract ("the so-called Spanish plume")
Categorisation of synoptic environments associated with mesoscale convective systems over the UK	Lewis, M.W.; Gray, S.L.	2010	<i>Atmospheric Research</i> 97(1-2):194–213	10.1016/j.atmosres.2010.04.001	ScienceDirect	Abstract + body ("classical/modified Spanish plume")
Improvements in nowcasting capability: analysis of three structurally distinct severe thunderstorms across northern England on 1 July 2015	Lewis, M.W.; Silkstone, N.	2017	<i>Weather</i> 72(4):91–98	10.1002/wea.2837	Wiley/RMetS	Body ("classic Spanish plume")
A climatology of convective available potential energy in Great Britain	Holley, D.M.; Dorling, S.R.; Steele, C.J.; Earl, N.	2014	<i>International Journal of Climatology</i> 34(14):3811–3824	10.1002/joc.3976	Wiley/RMetS	Abstract + body ("Spanish plume events")

Title	Authors	Year	Journal	DOI	Found via	Phrase location
Thunderstorm tracking in Northwest Europe for enhanced hazard preparedness	Hayward, L.; Whitworth, M.; Pepin, N.; Dorling, S.	2023	<i>International Journal of Climatology</i> 43(11):4894–4916	10.1002/joc.8123	Wiley/RMetS	Body ("the Spanish Plume weather pattern"); online 30 May 2023 Royal Meteorological Society

3. Unverified records (topically Spanish-plume; phrase in body NOT confirmable due to paywall)

Title	Authors	Year	Journal	DOI	Status
Europe extreme heat 22–26 July 2019: Was it caused by subsidence or advection?	de Villiers, M.P.	2020	<i>Weather</i> 75(8):228–235	10.1002/wea.3717	Phrase NOT in abstract; body inaccessible — UNVERIFIED
Severe thunderstorms over south-east England on 24 June 1994: A forecasting perspective	Young, M.V.	1995	<i>Weather</i> 50(7):250–256	10.1002/j.1477-8696.1995.tb06121.x	Phrase confirmed only in reference list (Morris 1986 citation); body inaccessible — UNVERIFIED
The synoptic setting of thunderstorms in western Europe	van Delden, A.	2001	<i>Atmospheric Research</i> 56(1–4):89–110	10.1016/S0169-8095(00)00092-2	Cited by others as a term source; body inaccessible — UNVERIFIED

4. Borderline / near-miss list (excluded, kept auditable)

- **Morris, R.M. (1986), "The Spanish plume – testing the forecaster's nerve," *Meteorological Magazine* 115:349–357** — exact phrase in title, but *Meteorological Magazine* is a Met Office professional magazine, not a peer-reviewed journal → excluded (non-peer-reviewed). Judgement call flagged. (Citation confirmed across multiple RMetS/AMS reference lists.)
- **Gray, M.E.B.; Marshall, C. (1998), "Mesoscale convective systems over the UK, 1981–97," *Weather*** — Spanish-plume-adjacent; exact phrase not confirmed in body → excluded pending verification.
- **Hernández-Ceballos et al. (2016), *Advances in Meteorology*** — heat-wave air-mass study over Spain; does not use the exact phrase → excluded (near-miss, topical only).

5. Excluded by date (post-13 Aug 2023) though phrase-matching

- **Schultz, D., Herrerias Azcue, F., Lowe, D., Kirshbaum, D.J. & Younge, M.V. (2025), "Rethinking the Spanish plume: An airstream analysis challenges the canonical conceptual model," *Quarterly Journal of the Royal Meteorological Society* 151(773):e5070, 10.1002/qj.5070** — Wiley records "First published: 06 August 2025"; also the designated non-independent source.
- **Schultz, D.M., Young, M.V. & Kirshbaum, D.J. (2025), "The Spanish Plume Elevated Mixed Layer: A Review of Its Use and Misuse within the Scientific Literature," *Monthly Weather Review* 153(5):737–761, 10.1175/MWR-D-24-0139.1** — 2025; designated non-independent source. Its abstract states verbatim: "A literature review of 102 peer-reviewed journal articles mentioning the Spanish plume is performed (of which 84 have only passing mentions)."
- "Atmospheric patterns associated with summer sub-daily rainfall extremes in western Europe," *Climate Dynamics* (2024), 10.1007/s00382-024-07440-7 — uses the phrase but 2024. Springer
- Ye et al. (2024), *International Journal of Climatology*, 10.1002/joc.8473 — 2024.

6. PRISMA-style flow account

- **Records retrieved before screening:** Not a verifiable numeric total, because none of the four databases returned an auditable count. Distinct candidate records surfaced through federated queries: ~13 unique journal-article records plus numerous non-scholarly hits (Wikipedia, Met Office, ZAMG/EUMETSAT training pages, news/blogs).
- **After deduplication (match on DOI/title+author+year):** 13 unique scholarly records carried forward (7 confirmed eligible + 3 unverified + 3 borderline/near-miss), with 4 additional phrase-matching records excluded on date.
- **After full screening — eligible: 7 confirmed** (with 3 additional "unverified" retained separately).
- **Exclusion reasons with counts:**
 - Post-cut-off date (>13 Aug 2023): 4 (2 Schultz reviews + *Climate Dynamics* 2024 + Ye 2024)
 - Non-peer-reviewed (magazine/technical): 1 (Morris 1986)
 - Phrase not confirmed present (near-miss/topical only): 2 (Gray & Marshall 1998; Hernández-Ceballos et al. 2016)
 - Full text inaccessible → phrase location unverifiable: 3 (de Villiers 2020; Young 1995; van Delden 2001)
 - Non-scholarly/non-peer-reviewed web material (encyclopedias, forecaster bulletins, news, training manuals): numerous, all excluded.

Recommendations

1. **To finalize the count, obtain authenticated access** to Web of Science (`(TS="Spanish plume")`), Scopus, Google Scholar, AMS Journals Online, and Wiley Online Library, and record verbatim query strings and raw counts. The current "unverified" database counts are the single biggest gap.
2. **Resolve the three unverified papers** (de Villiers 2020; Young 1995; van Delden 2001) by inspecting the full-text PDFs directly; if the exact phrase is absent from title/abstract/body, reclassify them to the near-miss list. For Young (1995) the phrase was confirmed only in the reference list (its citation of Morris 1986), which does not meet the inclusion criterion.
3. **Confirm Hayward et al. (2023) eligibility on date** — the authoritative early-online date is 30 May 2023 (before the 13 Aug 2023 cut-off); the print issue (Sept 2023, vol. 43 issue 11) post-dates the cut-off, so document that inclusion rests on the early-online date.
4. **Benchmark that would change conclusions:** if authenticated database searches surface additional *pre-2023* peer-reviewed articles with the exact phrase — plausible given the Schultz et al. (2025) *Monthly Weather Review* review's stated tally of "102 peer-reviewed journal articles mentioning the Spanish plume (of which 84 have only passing mentions)" — the eligible count should rise. Treat 7 as a **lower bound** from independent federated discovery, not a complete census. (Note: that 102-article figure is cited only as a magnitude sanity-check; per the independence constraint, no papers were harvested from that review.)

Caveats

- **Independence limitation (material):** The two Schultz et al. (2025) reviews saturated search results for nearly every query — including the historical framing that "the term originates from the Spanish-plume airstream, first discussed by Carlson and Ludlam (1968)." Every candidate paper therefore had to be independently located and verified on its own publisher page rather than harvested from those reviews' reference lists. The reviews' prominence structurally biases what a phrase search surfaces first and makes true independence difficult; this is documented rather than eliminated.
- **Count verifiability:** No native database returned an auditable raw count via available tools; all four counts are "unverified." The report therefore cannot give "records before deduplication" as a hard number.

- **Full-text access:** Wiley, ScienceDirect, and AMS full texts were frequently paywalled or bot-blocked, so several phrase confirmations rest on abstracts or high-confidence secondary evidence; three papers remain explicitly "unverified" rather than silently included/excluded.
- **Scope:** All confirmed-eligible papers concern European (chiefly UK/western-European) severe-convective or CAPE climatology contexts, consistent with the atmospheric sense of "Spanish plume"; no eligible paper used the term in a non-meteorological sense.